

Delivery Performance, Delay Risk, and Logistics Efficiency Analysis in Global Supply Chain Operations

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Abstract: This study evaluates delivery performance, delay risk, and logistics efficiency across APL Logistics' global fulfillment network. Using 180,519 validated shipment records spanning 164 destination countries, 23 order regions, 5 markets, and 4 shipping modes, the analysis measures the gap between actual and scheduled shipping duration and examines late-delivery risk across operational segments. The methodology combines delivery gap calculation, on-time classification, shipping mode efficiency indexing, and dimensional diagnostics for region, market, and customer segment performance. Results show that only 42.7% of shipments met or outperformed the scheduled timeline, while 57.3% were delayed and the mean delivery gap was 0.57 days. Standard Class achieved the strongest results despite the largest shipment volume, whereas First Class and Second Class displayed materially higher delay exposure. Geographic diagnostics identified Central Africa as the highest-risk region and Europe as the weakest market in the current dataset. The study concludes with recommendations for route governance, shipping-mode redesign, and stronger source-system data capture.

Keywords: ~~Index Terms~~ business intelligence, delay risk, logistics efficiency, shipping analytics, supply chain.

I. INTRODUCTION

Delivery reliability is a core operational requirement in global supply chains because shipping delays increase service-level risk, raise exception-handling cost, and weaken customer trust. APL Logistics operates across multiple regions, markets, and transportation modes, which makes delivery-performance monitoring a necessary part of day-to-day decision support. Without a clear view of late-delivery risk, operations teams are forced into reactive escalation instead of preventive logistics control.

This study investigates how actual shipping duration compares with scheduled shipping commitments, how delay exposure varies by shipping mode and geography, and whether specific customer segments face elevated service risk. The objective is to convert operational order data into an evidence-based logistics diagnostic that can support route review, service redesign, and performance governance.

II. DATASET AND METHODOLOGY

The analysis uses the APL Logistics operational dataset containing 40 business and shipment variables. After validation, 180,519 records were retained for analysis. The data spans 164 destination countries, 23 order regions, 5 markets, 3 customer segments, and 4 shipping modes: First Class, Same Day, Second Class, and Standard Class. Core analytical fields include actual shipping days, scheduled shipping days, delivery status, late-delivery risk, market, region, customer segment, and destination geography.

Data preparation focused on numeric conversion, consistency checks for shipping-duration fields, and validation of the binary late-delivery-risk indicator. A delivery delay gap was then computed as the difference between actual shipping days and scheduled shipping days.

Deliveries were classified as early, on-time, or delayed; on-time delivery rate and late-delivery risk ratio were calculated; and a shipping mode efficiency index was derived from the ratio of scheduled time to actual time. Aggregated diagnostics were produced for shipping mode, region, market, and customer segment. The source dataset does not contain order-date or shipment-date timestamp columns, so longitudinal trend and seasonality analysis could not be performed in this study.

III. RESULTS AND DISCUSSION

The overall delivery baseline indicates significant service pressure. Only 42.7% of validated shipments were delivered on time or early, while 57.3% were delayed. The average delay gap across the network was 0.57 days, and the late-delivery-risk ratio reached 54.8%. These results suggest that delay exposure is not limited to isolated exceptions but reflects a broad operational pattern that deserves managerial attention.

Shipping-mode analysis revealed strong performance dispersion. Standard Class carried the largest volume of shipments and still achieved the best service outcome, with a 60.2% on-time rate and an efficiency index of 89.4. Same Day performed moderately, with a 52.2% on-time rate. By contrast, Second Class showed a 76.6% late-delivery-risk ratio and an average delay gap of 1.99 days, while First Class exhibited the weakest outcome in the dataset, with a 95.3% late-delivery-risk ratio and no on-time shipments recorded in the current data extract. This result indicates a probable mismatch between premium service intent and actual fulfillment performance.

Regional and market diagnostics also identified meaningful hotspots. Central Africa recorded the highest regional delay exposure, with a 58.0% late-delivery-risk ratio and an average delay gap of 0.64 days. South Asia and Western Europe also showed elevated risk levels. At the market level, Europe delivered the weakest outcome, with a 55.2% late-delivery-risk ratio. Customer-segment differences were comparatively narrow, but Home Office customers experienced the highest risk at 55.1%,

implying that the delay problem is primarily network-wide rather than segment-specific.

IV. RECOMMENDATIONS

APL Logistics should prioritize operational review of First Class and Second Class fulfillment because both modes underperform relative to service expectations. Route-level improvement programs should focus first on Central Africa, South Asia, and Western Europe, while Europe should be treated as the most urgent market for logistics-efficiency review. Standard Class practices should be benchmarked and selectively transferred to weaker modes where feasible. Finally, the source system should be expanded to include real date-related columns for order creation, dispatch, and shipment completion so that future analysis can support trend, seasonality, and SLA-aging diagnostics.

V. CONCLUSION

This research demonstrates that delivery-performance analytics can provide a practical control layer for global logistics operations. The Project 4 analysis shows that delay exposure at APL Logistics is both structural and measurable, with meaningful differences across shipping modes and regions. By operationalizing the delivery gap, late-delivery risk, and logistics-efficiency indicators in a dashboard and research-driven reporting workflow, the organization gains a stronger basis for corrective action, service redesign, and future supply-chain optimization.

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